

Low cost GAN depth estimation for 3D object modeling

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Abstract—Depth estimation, one of the leading research in computer vision, from autonomous driving to 3D modeling, has play a key role to artificial perception of the machine. This paper present reconstruction of indoor 3D plane focusing on optimal cost to accuracy ratio, using simple Generative Adversarial Networks (GANs) trained using NYU Depth v2 dataset for indoor plane. Using this limitation our approach in lowering cost for optimal cost to accuracy ratio has achieved xx% accuracy using only xx training time resulted in xx times yield compared to other approach.

Index Terms—Depth Estimation, Generative Adversarial Network, 3D Modeling

I. Introduction

Monocular depth estimation (MDE) a challenging ill-posed problem due to inherit loss of geometric information when projecting 3D scene to 2D images. MDE research is one of the leading research topics from robotic, autonomous driving, and 3d modeling, with varied result but the overall accuracy is consistently high. Recent research on MDE heavily dominated by deep learning (DL) approach [4], focused on multiple DL architecture such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), variational auto-encoders (VAEs) and generative adversarial networks (GANs) [5].

Despite the success of deep learning (DL) based regression method in these high accuracy research

II. Related Work

Related work goes here.

III. Proposed Method

Proposed method on research.

IV. Experiment

Experiment details.

V. Discussion

Discussion about the results, and ablation study.

VI. Conclusion

Conclusion goes here.

VII. Acknowledgments

Aknowledgments goes here.

References

- [1] J. Chang and G. Wetzstein, “Deep optics for monocular depth estimation and 3d object detection,” in 2019 IEEE/CVF International Conference on Computer Vision (ICCV), 2019, pp. 10 192–10 201.
- [2] V. Guizilini, R. Ambruş, S. Pillai, A. Raventos, and A. Gaidon, “3d packing for self-supervised monocular depth estimation,” in 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 2482–2491.
- [3] A. Hendra and Y. Kanazawa, “Tp-gan: Simple adversarial network with additional player for dense depth image estimation,” *IEEE Access*, vol. 11, pp. 44 176–44 191, 2023.
- [4] A. Masoumian, H. A. Rashwan, J. Cristiano, M. S. Asif, and D. Puig, “Monocular depth estimation using deep learning: A review,” *Sensors*, vol. 22, no. 14, 2022. [Online]. Available: <https://www.mdpi.com/1424-8220/22/14/5353>
- [5] C. Zhao, Q. Sun, C. Zhang, Y. Tang, and F. Qian, “Monocular depth estimation based on deep learning: An overview,” *Science China Technological Sciences*, vol. 63, no. 9, p. 1612–1627, Jun. 2020. [Online]. Available: <http://dx.doi.org/10.1007/s11431-020-1582-8>
- [6] D. Eigen, C. Puhrsch, and R. Fergus, “Depth map prediction from a single image using a multi-scale deep network,” 2014. [Online]. Available: <https://arxiv.org/abs/1406.2283>